(v) $\left(e^{y}+1\right)\cos x\ dx+e^{y}\sin x\ dy=0$

(vi) $\left(ydx-xdy\right)\cot\left(\frac{x}{y}\right)=ny^{2}dx$

(vii) $\frac{dy}{dx}-x\sqrt{25-x^{2}}=0$

(viii) $x\cos y\ dy=e^{x}\left(x\log x+1\right)dx$

(ix) $\tan y\frac{dy}{dx}=\cos\left(x+y\right)+\cos\left(x-y\right)$

(x) $\frac{dy}{dx}=\tan^{2}\left(x+y\right)$

10.6.3 Homogeneous Form or Homogeneous Differential Equation

Definition 10.12 : (Homogeneous Function of degree *n*)

A function $f(x,y)$ is said to be a **homogeneous** function of degree n in the variables x and y if, $f(tx,ty)=t^{n}f(x,y)$ for some $n\in\mathbb{R}$ for all suitably restricted x,y and t. This is known as **Euler's homogeneity.**

For instance,

(i) $f(x,y)=6x^{2}+2xy+4y^{2}$ is a homogeneous function in x and y, of degree two.

(ii) But $f(x,y)=x^{3}+\left(\sin x\right)e^{y}$ is not a homogeneous function.

If $f(x,y)$ is a **homogeneous** function of degree zero, then there exists a function g such that $f(x,y)$ is always expressed in the form $g\left(\frac{y}{x}\right)$ or $g\left(\frac{x}{y}\right)$.

Definition 10.13: (Homogeneous Differential Equation)

An ordinary differential equation is said to be in **homogeneous form**, if the differential equation is written as $\frac{dy}{dx}=g\left(\frac{y}{x}\right)$.

Caution

The word "homogeneous" used in Definition 10.7 is different from in Definition 10.12.

Remark

(i) The differential equation $M(x,y)dx+N(x,y)dy=0$ [in differential form] is said to be **homogeneous** if M and N are **homogeneous functions of the same degree**.

(ii) The above equation is also written as $\frac{dy}{dx}=f(x,y)$ [in derivative form] where $f(x,y)=-M(x,y)/N(x,y)$ is clearly homogeneous of degree 0.

For instance

(1) consider the differential equation $\left(x^{2}-3y^{2}\right)dx+2xy\ dy=0$. The given equation is rewritten as $\frac{dy}{dx}=\frac{3y^{2}-x^{2}}{2xy}=\frac{3}{2}\left(\frac{y}{x}\right)-\frac{1}{2}\left(\frac{1}{y/x}\right)$. Thus, the given equation is expressed as $\frac{dy}{dx}=\frac{3}{2}\left(\frac{y}{x}\right)-\frac{1}{2}\left(\frac{1}{y/x}\right)=g\left(\frac{y}{x}\right)$. Hence, $\left(x^{2}-3y^{2}\right)dx+2xy\ dy=0$ is a homogeneous differential equation.

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